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Optimizing the growth procedures for CdSe crystal by thermal analysis techniques

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ABSTRACT

The growth procedures for CdSe crystal have been optimized by thermal analysis techniques and a high-quality single crystal with large size has been grown. CdSe crystal was determined to exhibit smooth interface growth based on Jackson's model; the growth rate equation in the vapor phase method was built and the thermodynamic parameters (activation energy and rate constant pre-factor) in the growth equation were obtained by thermogravimetic analysis of the sublimation process of the crystal. The growth rate distribution curves under various temperature differences between the source and growth region of growth ampoule were obtained, and then the growth procedures such as temperature difference, growth temperature and pulling rate were optimized. Using the optimizing vapor phase growth technique, a single crystal with 26 mm diameter and 45 mm length has been grown, and it was characterized by X-ray diffraction, scanning electron microscopy, energy dispersive analyzer of X-ray and IR spectrophotometers. The results show that the as-grown CdSe ingot has prefect crystallinity, good homogeneity and high transmittance, which showed that the vapor phase growth technique optimized by thermal analysis was a promising growth method for high-quality CdSe single crystal.

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1. Introduction

CdSe crystal is one of the II–VI semiconductor compounds with a relatively wide transparency range, large nonlinear optical coefficients, low optical losses [1] and appropriate birefringence [2,3]. At normal temperature and pressure, it crystallizes in a hexagonal close packed wurtzite structure [4] and belongs to the $P6_3mc$ (C_{6V}^4) space group [5]. The CdSe single crystal has captured significant interest due to its high potential for room temperature nuclear radiation detectors [6,7] and nonlinear optical devices [8–13]. Many methods, such as the temperature gradient solution zoning technique (TGSZ) [14], chemical vapor transport method (CVTM) [15], solution growth [16], melt-growth [17] and solid-state re-crystallization (SSR) [18], etc., have been used for CdSe single crystal growth. However up till now, all methods on good quality CdSe single crystal have been very difficult, due to its high melting point (1512 K) [19], low heat conductivity (0.044 W/Cm K) and high equilibrium partial pressures [20], which will lead to many defects including precipitates, inclusions and poly-crystals in the as-grown crystals. In addition, the repeatability of the single crystal growth is poor. In order to overcome the above shortcomings, in recent years our laboratory has modified the growth technique [21] (i.e. vertical unseeded vapor phase growth with multi-step purification (VUVGMP)) and the experimental result showed that it was an effective growth technique. However, until now, only very limited information on VUVGMP methods for CdSe single crystals was available, and high-quality single crystals with big size are still difficult to grow. Thus, the growth technique is in need of further investigation.

Our present work is focused on optimizing the growth procedures of VUVGMP and growing high-quality CdSe single crystal. The thermodynamic properties of CdSe crystal have been studied, the growth mechanism was determined and thermodynamic parameters in the growth rate equation, such as the activation energy and the rate constant pre-factor, have been tested by thermogravimetric analysis (TGA). The growth and heat treatment parameters and pulling rates were optimized from thermal analysis. A crack-free CdSe single crystal with Ø26 × 45 mm was grown by optimizing the VUVGMP method. The as-grown crystal was characterized using X-ray diffraction (XRD), scanning electron microscopy (SEM), energy dispersive analyzer of X-ray (EDAX), and IR and UV spectrophotometers.

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2. Thermal analysis

2.1. Growth mechanism and rate

According to Jackson's theory [22–23], the roughness of crystal interfaces could be determined by the following factor:

$$\alpha = \frac{L}{kT}\frac{\eta}{Z} \tag{1}$$

where L/kT was the entropy of formation (fusion), T was the transition temperature, and Z and η were the coordination numbers within the crystal and the plane, respectively.

From Clapeyron equation, we can get

$$\ln P = -\frac{L}{RT} + A \tag{2}$$

where R is the gas constant. The saturated vapor pressure of CdSe crystal can be obtained as [24]

$$\ln P_{\mathrm{CdSe}} = 14.823 - \frac{11067}{T} \tag{3}$$

The crystal growth temperature T_e can be chosen to be 1373 K, so $L/R = 11\,067$, and for CdSe crystal, $\eta = 4$ or 5 and $Z = 12$. Thus, $\alpha \geq 2.68$ exhibited that the CdSe crystal belonged to smooth interface growth.

For vapor deposition, the growth rate could be written as follows [25]:

$$R = A(\Delta P)^2 \tag{4}$$

where ΔP was the saturation vapor pressure difference between the source and growth region of growth ampoule, and A was the rate constant of the crystallization, which could be expressed as

$$A = A_C \exp\left(\frac{-E_C}{RT}\right) \tag{5}$$

Here A_C was the rate constant pre-factor and E_C was the activation energy of the crystallization.

As both sublimation and crystallization process of the crystal can be approximately treated as a mutually reversible process, the crystallization rate constants could be derived through the crystal sublimation process. The TGA of the sublimation process of the crystal can be written as

$$-\frac{dW}{dt} = \rho S R' \tag{6}$$

where R' is the crystal sublimation rate, $\rho = 5.81$ g/cm^3 [26] is the CdSe density and S is the surface of the CdSe cube, then form parameter η_1 (for cube, $\eta_1 = 6$) is imported and $S = \eta_1 V^{2/3}$. From Eqs. (4) and (6), we have

$$-\frac{dW}{dt} = \eta_1 A \rho^{1/3} W^{2/3} (-\Delta P)^2 \tag{7}$$

Here ΔP is the pressure difference in the surface of CdSe cube and it can be expressed as

$$\Delta P = P_{\mathrm{CdSe}} + cV_N \tag{8}$$

With the integral transform of the Clapeyron equation

$$P_{\mathrm{CdSe}} = P_0 \exp\left(-\frac{L}{RT}\right) \tag{9}$$

Here, $\ln P_0 = 14.82$ [24], using Eqs. (5)–(8), we obtain

$$\ln\left[\beta\left(-\frac{dW}{dT}\right)\right] = -\frac{E_C + 2L}{RT} + f(W) \tag{10}$$

in which

$$f(w) = \frac{2}{3}\ln W + D \tag{11}$$

$$D = \ln\eta_1 + \ln A_0 + 2\ln P_0 + \frac{1}{3}\ln\rho + 2\ln\left(1 + \frac{cV}{P_{\mathrm{CdSe}}}\right) \tag{12}$$

where β is the heating rate and V is the gas flowing rate [27].

2.2. TGA

Thermogravimetric analysis of CdSe crystal was performed using an SDT Q600 simultaneous DSC/TGA analyzer. Pure zinc was used for the temperature calibration. The CdSe crystal cubes were placed in an Al_2O_3 crucible under N_2 atmosphere. The plots of DTG curves in Fig. 1 were that the CdSe crystal cubes (200 mg) were heated to 1100 °C at a heat rate of 8, 10, 12, 14 and 16 °C/min and held there for 30 min with a N_2 gas rate of 100 ml/min.

Three straight lines were obtained by plotting $\ln[\beta(-dw/dT)]$ vs $-1/T$, as indicated in Fig. 2a–c. Then, the activation energy (E_C) for crystallization could be derived as 47.95 kJ mol^{-1}.

The linear fit of $f(W)$ vs $\ln(W)$ was shown in Fig. 3. The slope (0.66) agreed well with the theoretical value in Eq. (11) and the pre-factor of the crystallization was calculated to be 2.56×10^{-2} mm/day Pa^2.

2.3. Optimizing growth procedures

From Eqs. (4), (5) and (9), the crystal growth rate could be written as

$$R_C = A_C P_0^2 \exp\left(\frac{-E_C}{RT_C}\right)\left[\exp\left(\frac{-L}{RT_S}\right) - \exp\left(\frac{-L}{RT_C}\right)\right]^2 \tag{13}$$

where T_C and T_S are the temperatures of the source and growth zone of growth ampoule, respectively, and the temperature difference (ΔT) between two interfaces can be expressed as

$$\Delta T = T_S - T_C \tag{14}$$

The growth rate distribution curves under different ΔT (5, 10, 15, 20, 25, 30, 35 and 40 K) with the source temperature in the range 1100–1550 K are shown in Fig. 4. It is easy to see that regardless of whether ΔT is 5 or 40 K, when the source temperature is lesser than 1113 K, the crystal growth rate is always equal to zero. This result agrees well with the reports in our previous TGA experiment [27], which found that the sublimation point of CdSe is about 1113 K. With the same ΔT, the crystal growth rate increases exponentially with temperature. In the crystal growth experiment, considering both CdSe poly-crystals will partially decompose near its melting point and the capacity of growth

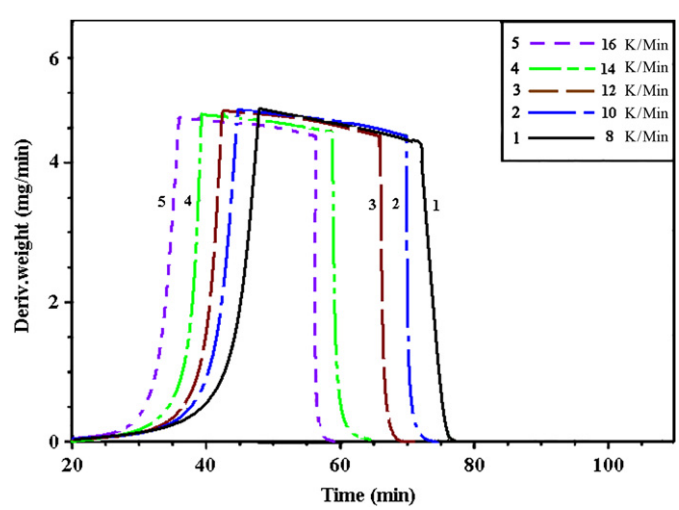

Fig. 1. DTG curves of pieces of CdSe crystal at different heating rates with N_2 flow rate 100 ml/min.

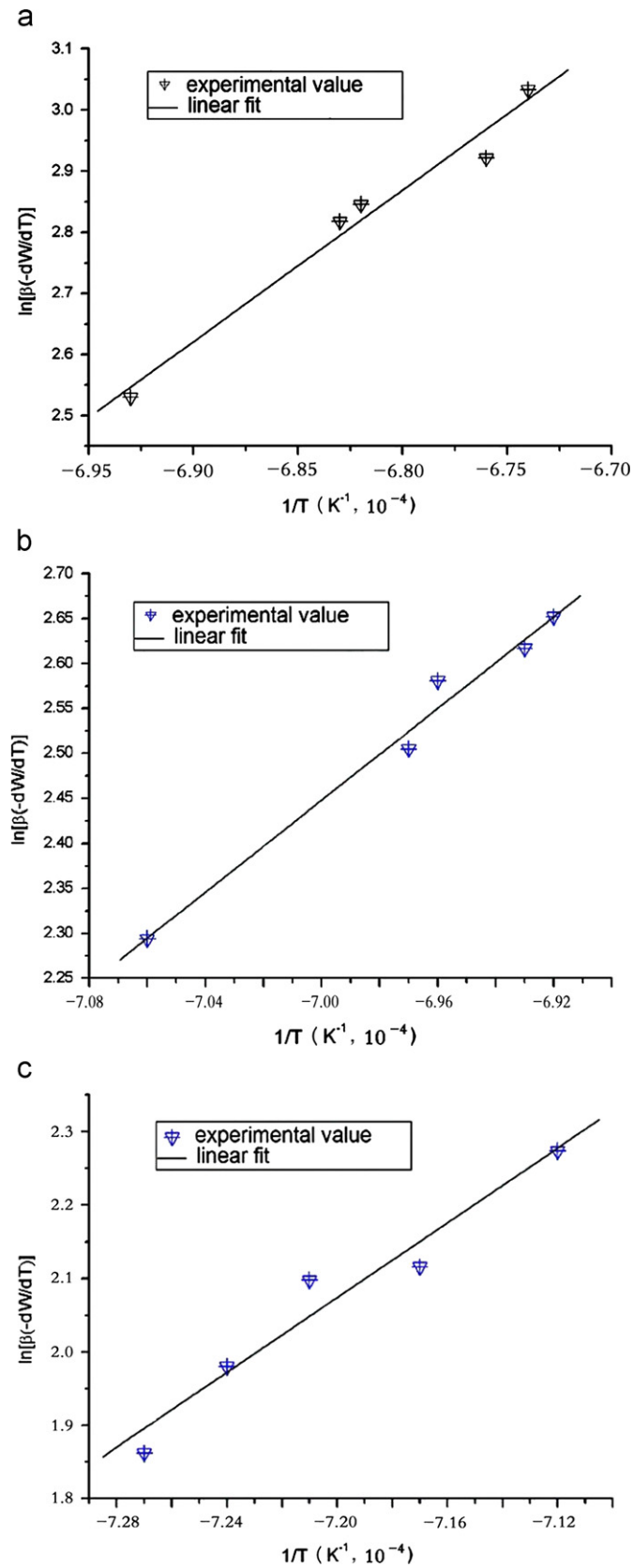

Fig. 2. Plots of dispersion $\ln[\beta(-dw/dT)]$ vs $1/T$ for CdSe crystal in different weights: (a) 50, (b) 100 and (c) 150 mg.

ampoule in high pressure and temperature for almost one month, the temperature of the source zone should be less than 1400 K. With a small fluctuation near this temperature, when ΔT is in the

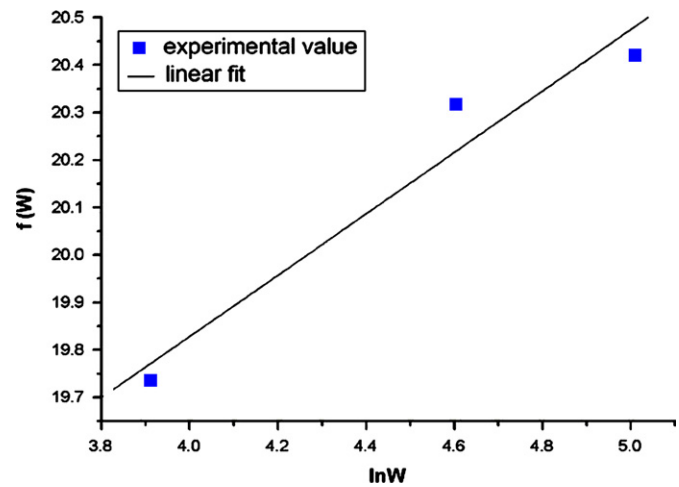

Fig. 3. $f(W)$ vs $\ln W$ plot for CdSe crystal cubes.

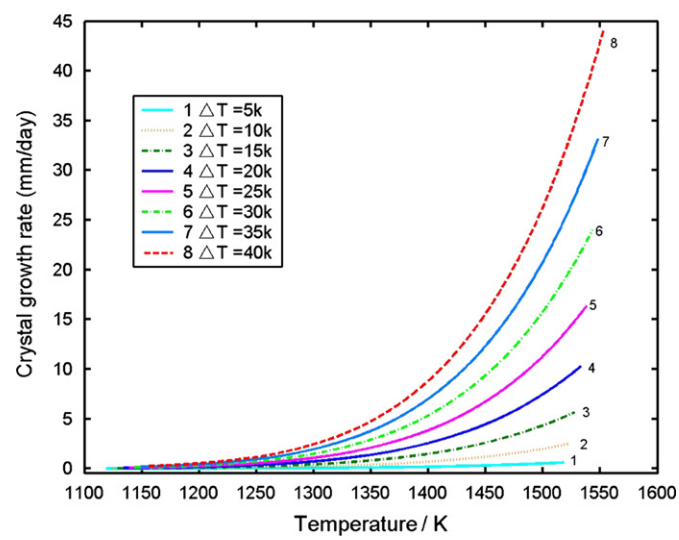

Fig. 4. Growth rate curves under different ΔT with source temperature in range 1100–1500 K.

scope of 20–25 K, the growth rate does not change obviously, whereas if it is larger than 25 K, the rate will have a significant change, which makes it difficult to grow steadily.

Therefore, for CdSe single crystal growth with the VUVGMP method, the temperature in the source zone of the growth ampoule should be less than 1400 K, and the temperature difference between the source and growth region should be kept at about 20–25 K. The crystal growth rate can be considered to be a constant when the temperature in both zones have been fixed, and the steady-state growth can be realized when the ampoule moves up with the growth rate.

3. Crystal growth and characterization

3.1. Crystal growth

Growth of the CdSe crystal was carried out in a vertical two-zone tubular electric furnace. The ampoule with CdSe polycrystalline was continuously evacuated under 6.2×10^{-4} Pa at 623 K for 3 h and then sealed. For the growth process, T_S and T_C were controlled at 1373 and 1348 K, respectively, and held there for 48 h to achieve big nucleation, and the ampoule was pulled at a rate of 2.8 mm/day

Fig. 5. Photograph of as-grown CdSe single crystal.

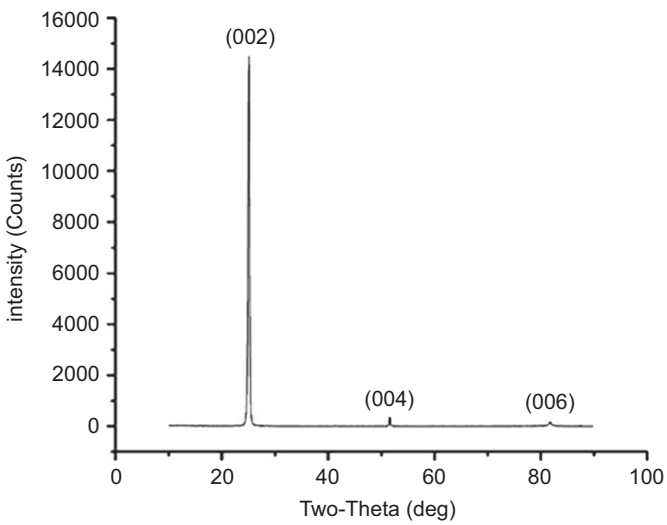

Fig. 6. XRD spectrum of {0 0 2} faces of CdSe crystal.

calculated by Eq.(13). After 6 weeks, the furnace was slowly cooled to 1040 K and held there for about 5 days for heat treatment, and then the ampoule was rapidly cooled to room temperature. The photograph of one of the as-grown CdSe crystals is shown in Fig. 5. As seen from the photograph, its size was up to Ø26 × 45 mm and the integrality was good.

3.2. X-ray diffraction analysis

Two cleavage faces were found in the ingot. XRD analysis showed they were {1 0 0} and (1 1 0) faces, respectively, which were consistent with the reported results by Siemens et al. [28]. What is more, the three-order diffraction peaks of the {0 0 2} faces with high intensity and good symmetry have been first presented in Fig. 6. The above-mentioned results demonstrated that the crystallinity of the as-grown CdSe ingot was good.

3.3. Observation and analysis on cleavage face

The shape of the cleavage face was observed by SEM and the image is shown in Fig. 7. As seen there were growth steps, which were parallel to each other and the intervals were about the same.

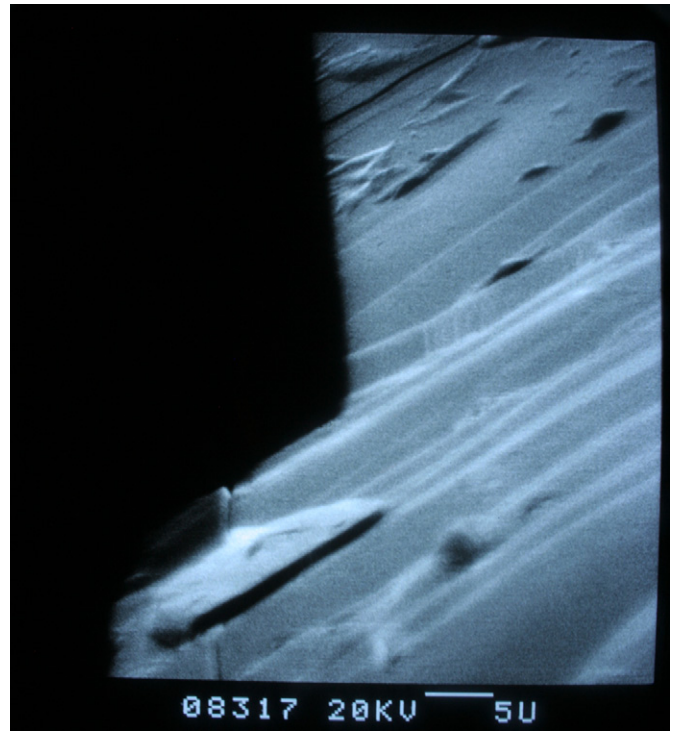

Fig. 7. SEM image of growth steps on cleavage faces.

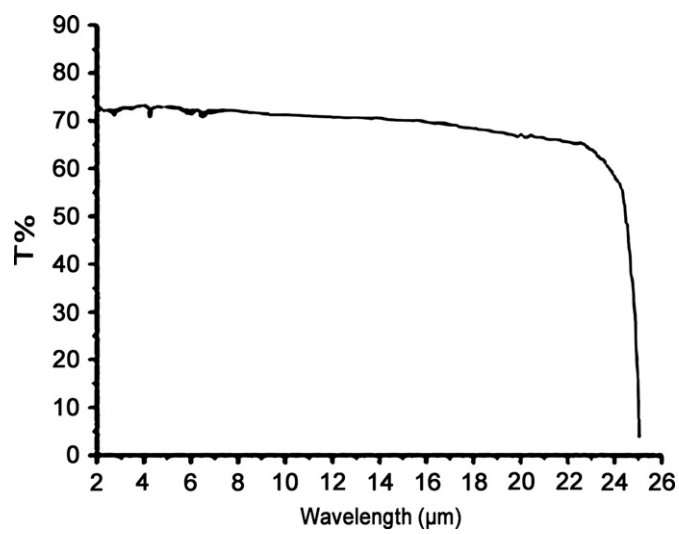

Fig. 8. IR transmission spectrum of as-grown CdSe crystal.

The stoichiometry ratio was tested by EDAX to be Cd:Se = 1:0.97, which was close to the ideal ratio of 1:1. These results demonstrated that the as-grown crystal was more homogeneous.

3.4. IR and vis–NIR spectrum tests

A wafer of 1.5 mm thickness was cut from the as-grown crystal of CdSe. The IR transmission spectrum recorded by an IRPrestige-21 spectrophotometer is shown in Fig. 8. It can be seen that the infrared transmission was above 70% in the region from 2 to 24 μm. The vis–NIR absorption spectrum recorded by a UV-1700 spectrophotometer is shown in Fig. 9. The result showed that the absorption edge of the as-grown crystal was nearby 730.2 nm and the band gap was calculated to be 1.70 eV.

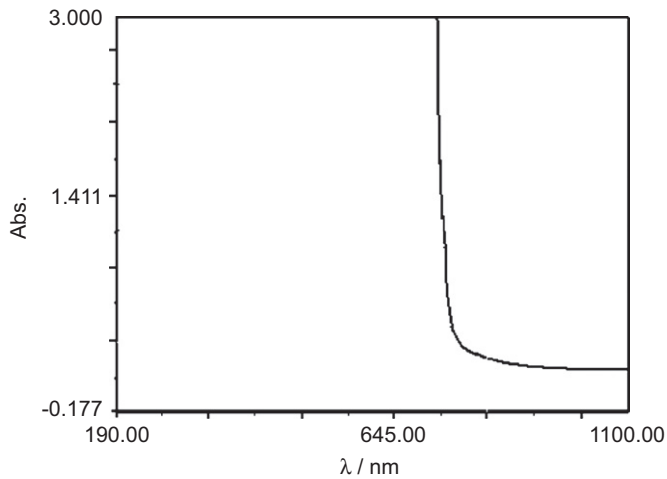

Fig. 9. Vis–NIR transmission spectrum of as-grown CdSe crystal.

4. Conclusions

The growth procedures for the CdSe crystal have been optimized by thermal analysis. The CdSe crystal was determined to exhibit smooth interface growth and the vapor phase growth rate equation was built. The activation energy was tested to be 47.95 kJ $mol^{-1}$and rate constant pre-factor was 2.56×10^{-2} mm/day Pa^2 by thermogravimetic analysis techniques. The growth rate distribution curves under various temperature differences between the source and growth region of the growth ampoule were drawn; the temperature in the source zone of the growth ampoule should be less than 1400 K and the temperature difference between the source and the growth region should kept at about 20–25 K. Then a crack-free CdSe single crystal 26 mm in diameter and 45 mm in length was grown by the optimized VUVGMP technique and it was characterized by different methods, such as XRD, SEM, IR, UV spectrophotometry, etc. The results show that the as-grown CdSe ingot has prefect crystallinity, good homogeneity and high transmittance, and can be used for the preparation of IR nonlinear optical devices.

The good quality of the as-grown crystals showed that the vapor phase growth technique optimized by thermal analysis perhaps was a promising method for big-size and high-quality CdSe single crystal growth.

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References

[1] K.L. Vodopyanov, J. Opt. Soc. Am. B 16 (1999) 1579.
[2] R.L. Herbst, R.L. Byer, Appl. Phys. Lett. 19 (1971) 527.
[3] W.L. Bond, J. Appl. Phys. 36 (1965) 1674.
[4] C.J. Anowitz, O. Gunther, G. Jungk, et al., Phys. Rev. B 5 (1994) 2181.
[5] S. Ninomiya, S. Adachi, J. Appl. Phys. 78 (1995) 4681.
[6] Y.V. Korostelin, V.I. Kozlovsky, A.S. Nasibov, et al., J. Cryst. Growth 159 (1996) 181.
[7] M. Roth, A. Burger, Appl. Phys. Lett. 52 (15) (1988) 1234.
[8] J.A. Weiss, L.S. Goldberg, Appl. Phys. Lett. 24 (1974) 389.
[9] R.G. Wenzel, G.P. Arnold, Appl. Opt. 15 (1976) 1322.
[10] A. Ferrario, M. Garbi, Opt. Commun. 17 (1976) 158.
[11] R.L. Herbst, R.L. Byer, Appl. Phys. Lett. 21 (1972) 189.
[12] D. Andreou, Opt. Commun. 23 (1977) 37.
[13] A.A. Davydov, L.A. Kulevskii, A.M. Prokhorov, et al., Opt. Commun. 9 (1973) 234.
[14] A. Burger, M. Roth, M. Schieber, IEEE Trans. Nucl. Sci. 32 (1985) 556.
[15] J.S. Yeong, S.J. Tae, K.S. Hyun, et al., New Phys. (Korean Phys. Soc.) 31 (1991) 784.
[16] M. Rubenstein, J. Cryst. Growth 3–4 (1968) 309.
[17] G. Goetz, A. Krostand, H.J. Schulz, J. Cryst. Growth 101 (1990) 414.
[18] R. Triboulet, J.O. Ndap, A.E. Mokri, et al., J. Phys. III 5 (1995) 141 (Suppl.).
[19] A. Burger, I. Shilo, M. Schieber, IEEE Trans. Nucl. Sci. 30 (1983) 368.
[20] R.C. Sharma, Y.A. Chang, J. Phase Equilibria 17 (1996) 140.
[21] S.F. Zhu, B.J. Zhao, Y.R. Jin, et al., J. Cryst. Growth 240 (2002) 454.
[22] K.A. Jackson, Liquid Metals and Solidification, ASM, Cleveland, 1958, p. 174.
[23] K.A. Jackson, J. Cryst. Growth 24 (1974) 130.
[24] Nauka, Physical–Chemical Properties of Semiconductors Handbook, Moscow, 1979.
[25] C.H.L. Goodman, Crystal Growth Theory and Techniques, Plenum Press, London, 1974, p. 117.
[26] K.J. Laidler, The World of Physical Chemistry, Oxford University Press, London, 1993.
[27] T.X. Zeng, B.J. Zhao, S.F. Zhu, J. Alloys Compd. 491 (2010) 170.
[28] B. Siemens, C.C. Domke, P. Ebert, et al., Phys. Rev. B. 59 (1999) 3000.